PV102 Day 4: Module Behavior String Design and Shade

Students connect PV101 series/parallel measurement habits to PV102 design judgment: string voltage, current, controller limits, temperature, and shade-sensitive layout decisions.

PV102 | CTS Day 20 | Instructor Run Sheet

String design follows limits, temperature, and shade reality. | Thursday, August 6, 2026

Core rule

String design follows limits, temperature, and shade reality.

Learning targets

- Predict how series and parallel wiring affect PV string voltage and current.
- Explain why controller input limits control string design.
- Recognize shade as both a production and mismatch problem.
- Use a safe paper/string model to compare design options before hardware work.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	PV module spec sheet samples String cards Calculator Shade cards Controller limit cards Whiteboard Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 3 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.
12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work

		triggers.
1:00-2:15	String Design Paper Lab	Teams compare three paper string options against controller limits and shade constraints.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 5 turns daily energy need into storage, autonomy, and battery-protection decisions.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use module behavior string design and shade to make the next design decision safer and more defensible. The rule is simple: String design follows limits, temperature, and shade reality."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Read the module and controller cards.
- Build three possible string layouts using cards.
- Calculate approximate operating voltage/current and open-circuit voltage.
- Mark shade exposure from Day 2 scenario notes.
- Reject any option that violates a limit or ignores shade evidence.
- Handoff the safest first-pass string layout and its required verification checks.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 5 turns daily energy need into storage, autonomy, and battery-protection decisions.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.

PV102 Day 4 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 20 | Morning Quiz - Answer Key
String design follows limits, temperature, and shade reality. | Thursday, August 6, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. In series wiring, matching module voltage primarily:

- A. Adds
- B. Disappears
- C. Becomes AC
- D. Turns into resistance

Answer: A

Rationale/look-for: Series wiring adds voltage while current is limited by the string.

2. In parallel wiring, matching branch current primarily:

- A. Adds
- B. Becomes zero
- C. Changes to frequency
- D. Removes voltage

Answer: A

Rationale/look-for: Parallel wiring adds current while voltage remains at branch voltage.

3. The charge controller input limit is important because:

- A. Exceeding limits can damage equipment or create unsafe conditions
- B. It chooses wire color
- C. It makes sunlight brighter
- D. It stores energy

Answer: A

Rationale/look-for: Equipment limits are hard design boundaries.

4. Shade on one module can affect:

- A. Only the paper worksheet
- B. Production and string behavior
- C. Lunch time only
- D. Nothing if the roof is new

Answer: B

Rationale/look-for: Shade reduces production and can create mismatch effects in connected circuits.

5. Cold temperature generally makes PV voltage:

- A. Higher
- B. Always zero
- C. AC only
- D. Unrelated to design

Answer: A

Rationale/look-for: Cold conditions can raise PV voltage and matter for maximum input limits.

6. Why should string design be checked before connecting equipment?

_____

_____

Answer: Safety and limits

Rationale/look-for: Students should mention voltage/current limits and equipment protection.

7. A controller allows 100 V max input. Should a design near 99 V be accepted without temperature check?

_____

_____

Answer: No

Rationale/look-for: Cold-weather voltage margin must be checked.

8. Name one reason parallel strings may require added protection.

_____

_____

Answer: Backfeed/current

Rationale/look-for: Parallel sources can increase fault current and may require OCPD depending on system design.

9. What evidence from Day 2 matters most for shade design?

_____

_____

Answer: Shade observations

Rationale/look-for: Shade timing, object location, and candidate array area.

10. Write the Day 4 rule in your own words.

_____

_____

Answer: Limits and shade

Rationale/look-for: Look for limits, temperature, shade, or electrical layout.

PV102 Day 4 Whiteboard Outline

PV102 | CTS Day 20 | Whiteboard Outline

String design follows limits, temperature, and shade reality. | Thursday, August 6, 2026

Board 1: Job	Board 2: Evidence	Board 3: Handoff
Topic: Module Behavior String Design and Shade Rule: String design follows limits, temperature, and shade reality. Question: What job are we solving?	Measured / Observed / Unknown Constraint that matters most Safety boundary before work	Decision Evidence Risk Unknowns Next action

Scenario anchors

- Four 22 Vmp / 27 Voc training modules feeding a 100 V controller.
- Two shaded morning modules and two unshaded modules on a shed roof.
- Long ground-rack wire run where voltage drop and current both matter.

Final board sentence

String design follows limits, temperature, and shade reality.

PV102 Day 4 Student Packet

Module Behavior String Design and Shade

Today's rule

String design follows limits, temperature, and shade reality.

Learning targets

- Predict how series and parallel wiring affect PV string voltage and current.
- Explain why controller input limits control string design.
- Recognize shade as both a production and mismatch problem.
- Use a safe paper/string model to compare design options before hardware work.

Morning quiz

1. In series wiring, matching module voltage primarily:

- A. Adds
- B. Disappears
- C. Becomes AC
- D. Turns into resistance

2. In parallel wiring, matching branch current primarily:

- A. Adds
- B. Becomes zero
- C. Changes to frequency
- D. Removes voltage

3. The charge controller input limit is important because:

- A. Exceeding limits can damage equipment or create unsafe conditions
- B. It chooses wire color
- C. It makes sunlight brighter
- D. It stores energy

4. Shade on one module can affect:

- A. Only the paper worksheet
- B. Production and string behavior
- C. Lunch time only
- D. Nothing if the roof is new

5. Cold temperature generally makes PV voltage:

- A. Higher
- B. Always zero
- C. AC only
- D. Unrelated to design

6. Why should string design be checked before connecting equipment?

_____

_____

7. A controller allows 100 V max input. Should a design near 99 V be accepted without temperature check?

_____

_____

8. Name one reason parallel strings may require added protection.

_____

_____

9. What evidence from Day 2 matters most for shade design?

_____

_____

10. Write the Day 4 rule in your own words.

_____

_____

Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

String Design Paper Lab

Teams compare three paper string options against controller limits and shade constraints.

Scenario options

- Four 22 Vmp / 27 Voc training modules feeding a 100 V controller.
- Two shaded morning modules and two unshaded modules on a shed roof.
- Long ground-rack wire run where voltage drop and current both matter.

Team procedure

- Read the module and controller cards.
- Build three possible string layouts using cards.
- Calculate approximate operating voltage/current and open-circuit voltage.
- Mark shade exposure from Day 2 scenario notes.
- Reject any option that violates a limit or ignores shade evidence.
- Handoff the safest first-pass string layout and its required verification checks.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

What made a string option unacceptable today: equipment limit, shade reality, or missing information?

_____

_____

PV102 Day 4 Afternoon Lab

String Design Paper Lab

PV102 | CTS Day 20 | Afternoon Lab
String design follows limits, temperature, and shade reality. | Thursday, August 6, 2026

Mission

Teams compare three paper string options against controller limits and shade constraints.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- Four 22 Vmp / 27 Voc training modules feeding a 100 V controller.
- Two shaded morning modules and two unshaded modules on a shed roof.
- Long ground-rack wire run where voltage drop and current both matter.

Procedure

- Read the module and controller cards.
- Build three possible string layouts using cards.
- Calculate approximate operating voltage/current and open-circuit voltage.
- Mark shade exposure from Day 2 scenario notes.
- Reject any option that violates a limit or ignores shade evidence.
- Handoff the safest first-pass string layout and its required verification checks.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	
Primary decision	

Evidence	
Safety boundary	
Unknowns	
Next safest action	

PV102 Day 4 Classroom Post

Copy-ready announcement and assignment text

PV102 | CTS Day 20 | Google Classroom Post
String design follows limits, temperature, and shade reality. | Thursday, August 6, 2026

Announcement

Today in PV102 we are working on module behavior string design and shade. The core rule is: String design follows limits, temperature, and shade reality.

Assignment title

PV102 Day 4 - Module Behavior String Design and Shade

Student directions

- Complete the morning readiness quiz independently.
- Use the reference section during the afternoon lab.
- Work with your team to complete the scenario handoff.
- Complete the exit ticket before class ends.

Turn in

- Completed student packet
- Team lab handoff
- Exit ticket

Success criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record unknowns.